

Heat Transfer Exam II

$$\tau_s = \mu \left. \frac{\partial u}{\partial y} \right|_{y=0} \quad C_f = \frac{\tau_f}{\rho u_\infty^2 / 2} (\text{local}) \quad Pr \triangleq \frac{\nu}{\alpha}$$

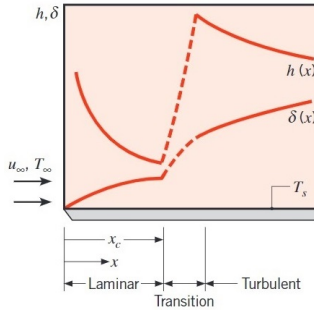
$$h \triangleq \frac{-k_f \left. \frac{\partial T}{\partial y} \right|_{y=0}}{T_s - T_\infty} \quad \bar{h} \triangleq \frac{1}{A_s} \int_{A_s} h dA_s \quad \bar{h}_{\text{plate}} = \frac{1}{L} \int_{A_s} h dx$$

$$Nu \triangleq \frac{hL}{k_f} = C Re_L^m Pr^n \quad Nu \triangleq f(Re_L, Pr)$$

$$T_f = \frac{T_s + T_\infty}{2} \quad C_D = \frac{F_D}{A_f(\rho u^2 / 2)} \quad A_f \equiv \text{frontal area}$$

External Flow: Flat Plate

$$Re_{\text{critical}} = \rho \frac{u_\infty x}{\mu} = \frac{u_\infty x}{\nu} = 5 \times 10^5$$



Mixed Boundary Layer

$$\bar{h}_L = \frac{1}{L} \left(\int_0^{x_c} h_{\text{lam}} dx + \int_{x_c}^L h_{\text{turb}} dx \right)$$

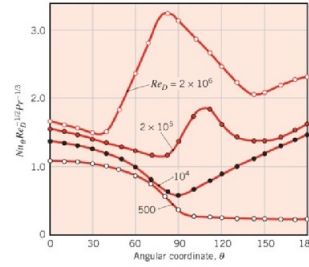
$$\overline{Nu}_L = (0.037 Re_L^{4/5} - A) Pr^{1/3} \quad A = 871$$

$$\Rightarrow \begin{cases} 0.6 \leq Pr \leq 60 \\ Re_x \leq Re_L \leq 10^8 \end{cases}$$

External Flow: Cylinder

$$Re_{\text{critical}} = \rho \frac{u_\infty D}{\mu} = \frac{u_\infty D}{\nu} = 2 \times 10^5$$

$$\theta_{\text{lam}} \approx 80 \text{ deg} \quad \theta_{\text{turb}} \approx 140 \text{ deg}$$



Re_D	C	m
0.4–4	0.989	0.330
4–40	0.911	0.385
40–4000	0.683	0.466
4000–40,000	0.193	0.618
40,000–400,000	0.027	0.805

$$Pr \geq 0.7 \quad \Rightarrow \quad \overline{Nu}_D = \frac{\bar{h}D}{k_f} = C Re_D^m Pr^{1/3}$$

External Flow: Sphere

$$Re_{\text{critical}} = \rho \frac{u_\infty D}{\mu} = \frac{u_\infty D}{\nu} = 2 \times 10^5$$

$$\overline{Nu}_D = 2 + (0.4 Re_D^{1/2} + 0.06 Re_D^{2/3}) Pr^{0.4} \left(\frac{\mu}{\mu_s} \right)^{1/4} = \frac{\bar{h}D}{k_f}$$

$$\Rightarrow \begin{cases} 0.71 \leq Pr \leq 380 \\ 3.5 \leq Re_D \leq 7.6 \times 10^4 \\ 1.0 \leq \mu/\mu_s \leq 3.2 \end{cases}$$

All properties evaluated at T_∞ , except μ_s is at T_s

Laminar Boundary Layer

$$\delta = \frac{5x}{\sqrt{Re_x}} \quad C_{f,x} = \frac{\tau_x}{\rho u_\infty / 2} = 0.664 Re_x^{-1/2}$$

$$Pr \geq 0.6 \Rightarrow \begin{cases} \frac{\delta}{\delta_T} \approx Pr^{1/3} \\ Nu_x = \frac{h_x x}{k_f} = 0.332 Re_x^{1/2} Pr^{1/3} \\ \overline{Nu}_x = \frac{\bar{h}_x x}{k_f} = 0.664 Re_x^{1/2} Pr^{1/3} \end{cases}$$

$$\overline{C}_{f,x} = 2C_{f,x} = 1.328 Re_x^{-1/2}$$

$$Pe_x \triangleq Re_x Pr$$

$$Pe_x \geq 100 \Rightarrow \begin{cases} Pr \leq 0.05 \Rightarrow Nu_x = 0.564 Pe_x^{1/2} \\ \text{all } Pr \Rightarrow Nu_x = \frac{0.3387 Re_x^{1/2} Pr^{1/3}}{[1 + (0.0468/Pr)^{2/3}]^{1/4}} \end{cases}$$

Turbulent Boundary Layer

$$Re_{x,crit} \leq Re_x \leq 10^8 \Rightarrow \begin{cases} C_{f,x} = 0.0529 Re_x^{-1/5} \\ \delta_T \approx \delta = 0.37 x Re_x^{-1/5} \end{cases}$$

$$0.6 \leq Pr \leq 60 \quad \Rightarrow \quad Nu_x = 0.0296 Re_x^{4/5} Pr^{1/3}$$

Internal Flow: Circular

$$Re_{\text{critical}} = \rho \frac{u_m D}{\mu} = \frac{u_m D}{\nu} = 2300$$

In fully developed region, $u \neq f(x)$, $h \neq f(x)$

$$Re_D = \frac{4\dot{m}}{\pi D \mu} \quad u_m \triangleq \frac{2}{r_o^2} \int_0^{r_o} u(r, x) r dr = \frac{\dot{m}}{\rho A_c}$$

$$\left(\frac{x_{fd,H}}{D} \right)_{\text{lam}} \approx 0.05 Re_D \quad \left(\frac{x_{fd,H}}{D} \right)_{\text{turb}} \approx 10$$

$$\left(\frac{x_{fd,T}}{D}\right)_{lam} \approx 0.05 Re_D Pr \quad \left(\frac{x_{fd,T}}{D}\right)_{turb} \approx 10$$

$$u(r) = \frac{1}{4\mu} \left(\frac{dP}{dx}\right) r_o^2 \left[1 - \left(\frac{r}{r_o}\right)^2\right]$$

$$\frac{u(r)}{u_m} = 2 \left[1 - \left(\frac{r}{r_o}\right)^2\right]$$

$$q = \dot{m}c_p(T_{m,o} - T_{m,i}) = \bar{h}A_s(T_s - T_m)$$

$$T_m = \frac{2}{u_m r_o^2} \int_0^{r_o} u T r dr \quad \bar{T}_m = \frac{T_{m,i} + T_{m,o}}{2}$$

$$\frac{dT_m}{dx} = \frac{P}{\dot{m}c_p} q_s'' = \frac{P}{\dot{m}c_p} h(T_s - T_m)$$

Reynolds Number

if $Re_D \leq 2300$,

$$\text{eval @ } \bar{T}_m \Rightarrow \begin{cases} q_s'' = \text{const} \Rightarrow Nu_D = \frac{hD}{k_f} = 4.36 \\ T_s = \text{const} \Rightarrow Nu_D = \frac{hD}{k_f} = 3.66 \end{cases}$$

else

$$Nu_D = 0.23 Re_D^{4/5} Pr^n \Rightarrow \begin{cases} n = 0.4 & T_s > T_m \\ n = 0.3 & T_s < T_m \end{cases}$$

$$\Rightarrow \begin{cases} 0.6 \leq Pr \leq 160 \\ Re_D \geq 10,000 \\ L/D \geq 10 \end{cases}$$

Constant Uniform q_s''

$$q_{conv} = q_s'' PL \quad T_m(x) = T_{m,i} + \frac{P}{\dot{m}c_p} q_s'' x$$

Constant Uniform T_s

$$q_{conv} = \dot{m}c_p(T_{m,o} - T_{m,i}) = \dot{m}c_p(\Delta T_i - \Delta T_o)$$

$$\Delta T_i \triangleq T_s - T_{m,i} \quad \Delta T_o \triangleq T_s - T_{m,o}$$

$$\Delta T_{lm} \triangleq \frac{\Delta T_o - \Delta T_i}{\ln(\Delta T_o / \Delta T_i)} \quad q_{conv} = \bar{h}A_s \Delta T_{lm}$$

Constant Uniform T_∞

$$\frac{\Delta T_o}{\Delta T_i} = \frac{T_\infty - T_{m,o}}{T_\infty - T_{m,i}} = \exp\left(-\frac{\bar{U}A_s}{\dot{m}c_p}\right)$$

$$\bar{U}A_s = \frac{1}{R_{tot}} \quad q_{conv} = \bar{U}A_s \Delta T_{lm}$$

Internal Flow: Non-Circular

$$T_m = \frac{1}{\dot{m}c_p} \int_{A_c} \rho U c_p T dA_c$$

$$D_h = \frac{4A_c}{P} \quad Re_D = \frac{\rho u_m D_h}{\mu}$$

if $Re_D \leq 2300$,

Cross Section	$\frac{b}{a}$	$Nu_D = \frac{hD_h}{k}$		Re_{D_h}
		(Uniform q_s'')	(Uniform T_s)	
	—	4.36	3.66	64
	1.0	3.61	2.98	57
	1.43	3.73	3.08	59
	2.0	4.12	3.39	62
	3.0	4.79	3.96	69
	4.0	5.33	4.44	73
	8.0	6.49	5.60	82
	∞	8.23	7.54	96
	∞	5.39	4.86	96
	—	3.11	2.49	53

else

Use the correlation listed under the Reynolds Number subsection.

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